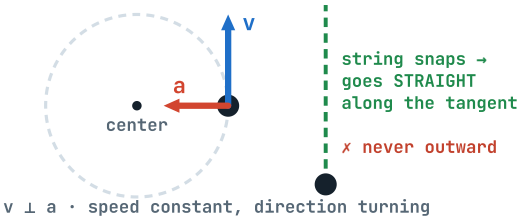
AP PHYSICS 1 · TOPIC 2.9 (+2.6 ORBITS)

Circular Motion on One Page

One idea runs all of it: **find the net force toward the center and set it equal to mv^2/r.** "Centripetal" is a job title, not a new force.

THE PICTURE TO REMEMBER



EQUATIONS: KNOW WHAT EACH ONE MEANS

$a_c = v^2/r$	points to the center · constant speed still accelerates
$\Sigma F_{in} = mv^2/r$	Newton's 2nd law along the radius: + toward the center
$v = 2\pi r/T$	one lap (circumference) per period T
$f = 1/T$	laps per second (Hz)
$a_c = 4\pi^2 r/T^2$	use this when you're given a period
$v = \sqrt{(GM/r)}$	circular orbit · r from the planet's center
$v_{min} = \sqrt{(gr)}$	top of a loop/bucket, when N or T = 0

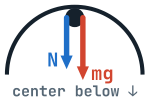
THE 4-STEP RECIPE (EVERY PROBLEM)

1. **FBD of real forces only** (gravity, normal, tension, friction). No F_c arrow.
2. **Find the center.** Choose + toward the center.
3. **$\Sigma F_{in} = mv^2/r$**: forces pointing in are +, out are –.
4. **Solve in symbols, then check a limit** (v→0? r→∞? does the sign make sense?).

WHO DOES THE CENTRIPETAL JOB?

SITUATION	REAL FORCE(S) POINTING IN
Ball on a string (horizontal)	Tension
Car on a flat curve	Static friction (sideways)
Banked curve, no friction	Horizontal part of the normal force
Satellite / Moon	Gravity
Top of a loop (inside)	Gravity + normal, both down
Spinning washer drum	Normal force from the wall

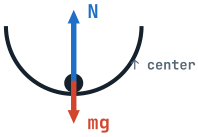
Top of a loop (inside)



$N + mg = mv^2/r$

Both forces point down (in). N = 0 → $v_{min} = \sqrt{(gr)}$.

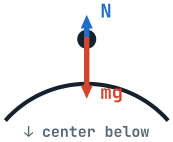
Bottom of a loop / dip



$N - mg = mv^2/r$

N > mg: you feel heavier. A ball on a string at the bottom: T – mg = mv^2/r.

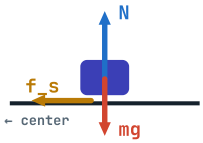
Hilltop (outside)



$mg - N = mv^2/r$

N < mg: lighter. N = 0 at $v = \sqrt{(gr)}$ → airtime.

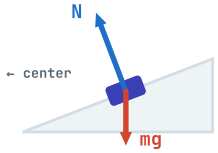
Flat curve (rear view)



$f_s = mv^2/r$

$v_{max} = \sqrt{(\mu_s gr)}$, and mass cancels. On ice the car goes straight.

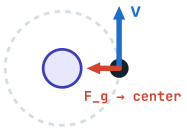
Banked curve, no friction



$N \sin\theta = mv^2/r \cdot N \cos\theta = mg$

Divide: $\tan\theta = v^2/(gr)$. There's one "design speed".

Circular orbit



$GMm/r^2 = mv^2/r \rightarrow v = \sqrt{(GM/r)}$

Gravity is the only force. Closer → faster, shorter T ($T^2 \propto r^3$).

SCALING: ASK "WHAT'S HELD CONSTANT?"

CHANGE	FIXED	F OR A
v × 2	r	**× 4**
v × 3	r	**× 9**
r × ½	v	**× 2**
r × 2	T (same ride)	**× 2**
orbit r × 4	planet	v × ½

MYTHS THAT COST POINTS

- ✗ **"Centrifugal force"**: you feel inertia, not an outward force.
- ✗ **Constant speed → a = 0**: the direction changes, so $a = v^2/r$.
- ✗ **Released → flies outward**: it goes along the tangent.
- ✗ **Tension points up at the top**: a string always pulls toward the center.
- ✗ **Astronauts have no gravity**: they're in free fall (about 0.9 g).
- ✗ **Heavier satellite → faster**: m cancels in $v = \sqrt{(GM/r)}$.

FRQ SENTENCES THAT SCORE

- "Taking toward the center as positive, **$\Sigma F = mv^2/r$** (Newton's 2nd law)."
- "At the top, gravity points **toward** the center; at the bottom it points **away**, so..."
- "**Just barely** / minimum speed → set F_N (or T) = 0."
- "The velocity changes **direction**, so the object accelerates toward the center."
- Proportions: **write the equation, then the ratio** ($F \propto v^2 \rightarrow 2^2 = 4$).